Response to Reviewers

Paper 1217, Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

We thank the reviewers and the SPC for their careful and constructive comments. We found the feedback very helpful in identifying several places where the original manuscript needed clearer definitions, additional validation, and more cautious interpretation. We have revised the manuscript substantially in response. All additions and major revisions are highlighted in blue in the revised manuscript.

The main changes include additional robustness checks for repost-cascade reconstruction, expanded validation of the ideology labels, clarification of the post-level unit of analysis, sensitivity analyses using alternative cascade and ideology specifications, and a revised regression specification after identifying a problem with the original scale estimator. We also expanded the descriptive reporting, motif analysis, and discussion of the scope and limitations of the findings.

Across these sensitivity analyses, estimates sometimes change in magnitude, but the substantive conclusions remain consistent. In particular, the cross-platform difference in reply-cascade structure persists under alternative samples and cascade representations, and ideological composition and interaction alignment remain strongly associated with the remaining platform difference under alternative ideology thresholds, label-noise simulations, and composition specifications. Some checks also narrow the interpretation of particular findings. For example, collapsing repeated posts by the same user reduces the breadth difference, indicating that repeat participation contributes to the post-level breadth contrast, while the partisan-only analysis shows that the residual center category contributes to part of the statistical attenuation. We have revised the manuscript to make these qualifications explicit.

#	Raised by	Reviewer point	Response	Where
C1	SPC-1, R1-R3	Repost cascades are reconstructed rather than directly observed, so the apparent cross-platform similarity could depend on the reconstruction method.	We agree that the reconstructed repost cascades require additional validation. We therefore rebuilt all 244,129 repost cascades under alternative linking rules, together with a 40-draw random-linking specification. Across these specifications, the platform-by-size interaction stays within [-0.012, +0.031] for breadth and [-0.083, +0.028] for depth, against +0.170 and -0.182 for the corresponding reply cascades. We also now refer to these cascades as reconstructed throughout the manuscript.	App. D, Fig. 9

#	Raised by	Reviewer point	Response	Where
C2	R3, SPC-1	A follower relationship observed after collection may not have existed at the time of reposting.	This is an important limitation because the follower networks are post-collection snapshots without edge creation dates. To examine the sensitivity of the results to this uncertainty, we randomly removed 5% to 30% of follower edges and reconstructed the cascades. The resulting platform interactions remain much smaller than those observed for reply cascades. Both follower networks were collected during the same period, although we do not assume that timing error is necessarily identical across platforms.	App. D
C3	SPC-2, R2, R3	It was unclear whether cascade nodes represent users or posts, and how repeated users are treated.	We have clarified that reply-cascade nodes are posts, not unique users. A user who replies multiple times therefore contributes multiple nodes carrying the same user-level ideology label. We also reconstructed the reply cascades after collapsing repeated appearances by the same user. The depth difference remains under this representation, whereas the breadth difference becomes substantially smaller. We now state explicitly that part of the post-level breadth contrast reflects repeat participation within threads.	Methods; App. E
C4	SPC-3, R1, R2	The ideology validation sample was limited and did not report class- or platform-specific performance.	We expanded the validation analysis to report class-specific and platform-specific precision and recall, with bootstrap confidence intervals, using the 171 cases on which the two human annotators agreed. Overall accuracy against the consensus labels is 0.86. Inter-annotator agreement is kappa = 0.77, and model-human agreement is kappa = 0.61 and 0.74. We also report the lower precision observed for the ideological minority class on each platform and its associated uncertainty.	App. B, Table 3
C5	SPC-3, R2	The center category may combine genuinely moderate users with mixed or uncertain classifications.	We agree that the center category should not be treated as a homogeneous group of ideological moderates. We now define it explicitly as a residual category that includes users who do not reach either partisan threshold. We also recomputed composition and alignment using partisan users only. The association with ideological structure becomes smaller but remains substantial, and we have narrowed the interpretation of the center category accordingly.	Methods; App. B

#	Raised by	Reviewer point	Response	Where
C6	SPC-3, R1	The manuscript should show whether the findings are robust to plausible ideology-label error.	We propagated the observed classification error through the full analysis rather than treating the ideology labels as error-free. For each error specification, user labels were perturbed, cascade-level composition and alignment were recomputed, and Model 3c was refit over 100 draws. We considered platform-specific errors, pooled errors, uncertainty in the estimated confusion matrix, and the less favorable performance measured against a single annotator. Across these simulations the median run retains 86 to 96 percent of the attenuation and the least favorable specification retains 70 percent, so the substantive conclusion is unchanged.	App. B
C7	R1	The estimated left share on Truth Social and right share on Bluesky appear high relative to previous estimates.	We agree that the ideological composition in our sample differs from platform-wide estimates reported in previous work. We now emphasize that our sampling frame consists of users participating in Biden- or Trump-related conversations during one campaign month rather than a representative sample of either platform. We also applied a prevalence correction based on the measured confusion matrix. This reduces the estimated Bluesky right share from 24.2% to 17.9% and the Truth Social left share from 11.5% to 1.9%. A remaining difference for Bluesky may partly reflect the candidate-centered sampling frame, and we now state this cautiously in the manuscript.	App. B
C8	SPC-4, R2, R3	Alignment is calculated from the same reply edges that determine cascade structure, making causal language inappropriate.	We agree and have revised the manuscript throughout to avoid a causal interpretation of the alignment results. Alignment and cascade geometry are jointly realized features of the same conversation, so the observed association does not establish a direction of influence. The Results now state this limitation directly, and the Discussion consistently describes composition and alignment as statistically associated with the platform difference rather than as causal mechanisms.	Results; Limitations
C9	SPC-5, R1	The paper needs more readable descriptive statistics, including the number of cascades at different sizes.	We added a descriptive table reporting cascade counts, post counts, and cascade-size distributions by platform and cascade type. This also made clear that root-only cascades are common and substantially more frequent on Bluesky than on Truth Social. Because these observations contain no variation in breadth or depth, we additionally refit the models after excluding size-1 cascades. The baseline platform difference becomes smaller, but remains present, and the combined ideology specification continues to account for most of the remaining difference.	App. C, Table 4; App. I, Table 6
C10	R1	Raw motif counts are needed in addition to standardized motif scores.	We added the observed counts for all 54 ideology-labeled three-node motifs on both platforms so that the standardized motif results can be interpreted alongside their absolute frequency.	App. G, Table 5

#	Raised by	Reviewer point	Response	Where
C11	R1	Motifs overlap, so the manuscript should explain whether the null model accommodates this dependence.	We now describe the motif randomization and counting procedure explicitly. The same overlapping-instance enumeration is applied to both the observed and randomized graphs, so overlap is treated consistently in both. We also clarify that neighboring motif statistics are correlated because motifs share substructure. Accordingly, we interpret the motif results as patterns across motif families rather than as independent pieces of evidence.	Methods; App. G
C12	SPC-6, R2	The main text describes robust regression with Huber loss, whereas the appendix referred to OLS with HC3 standard errors.	We corrected this inconsistency. The main reply-cascade models use the Huber loss. In reviewing the estimation procedure, we also found that the default median-absolute-deviation scale estimate degenerates because of the large mass of root-only cascades. We therefore refit the main models using Huber's Proposal 2 scale, which does not degenerate in these data. Table 1 now reports the corresponding estimates and standard errors, and Appendix I reports sensitivity results across samples and scale estimators. OLS with HC3 errors is used only for the repost-reconstruction robustness analysis, where the Huber scale estimate also degenerates.	Methods; App. I, Table 6
C13	SPC-7, R1-R3	The paper examines one month of Biden- and Trump-related discussion during an unusual election period. The paper should clarify what can be generalized and what the two cascade shapes imply.	We revised the Discussion to state more clearly both the substantive implications and the limits of the comparison. In particular, we distinguish the structural interpretation of broad/shallow and narrow/deep reply cascades from any normative evaluation of those forms. We also make clear that the observed ideological composition and absolute cascade statistics are specific to the candidate-centered sampling frame and period studied here.	Discussion; Limitations
C14	R1	The rationale for examining highly followed users only on Truth Social was not sufficiently developed, and a symmetric top-n rule might be preferable.	We expanded the justification for treating the unusually high-follower Truth Social accounts separately. The upper tail of the follower distribution is substantially more concentrated on Truth Social: the largest account has 56 times the follower count of the platform's 99.9th percentile account, compared with a ratio of 5 on Bluesky, and the top eight accounts account for a much larger share of total followers. We therefore treat the Truth Social group as a platform-specific concentration rather than imposing the same numerical cutoff on Bluesky, where there is no comparable separation in the upper tail.	App. A

#	Raised by	Reviewer point	Response	Where
C15	R1	How much of the observed reply depth consists of back-and-forth interaction between the same two users?	We added a direct analysis of A -> B -> A exchanges. Among cascades with depth of at least two, 68.8% on Bluesky and 70.0% on Truth Social contain at least one such exchange. These exchanges account for 20.0% and 15.6% of reply edges, respectively. The similar prevalence across the two platforms suggests that the observed depth difference is not primarily due to one platform containing more dyadic back-and-forth conversations.	App. F
C16	R1	More detail is needed about the BERTopic procedure and the treatment of outliers.	We expanded the topic-model description to report the corpus size (125,623 root posts), the HDBSCAN outlier reduction procedure and settings, the remaining outlier share, and the eleven topic categories used in the analysis.	App. H
C17	R1	The alignment results were difficult to follow.	We revised the alignment subsection to provide a clearer definition, define the notation at first use, and include a worked example showing how the measure is calculated within a reply tree.	Methods
C18	SPC-5, R1	Several figures were difficult to read because of overlapping labels and unclear plotted quantities.	We revised the figures to improve readability, including moving motif-family labels outside the plotting area and correcting labels and annotations that extended beyond the axes. The revised figures are rendered at 300 dpi and no longer contain overlapping elements.	Figs. 1, 4, 5, 7, 11
C19	R1	The literature review would benefit from more recent information-spreading work.	We updated the Related Work section with more recent work on cascade structure, information diffusion, and newer social media platforms, including recent work on Bluesky.	Related Work
C20	R1	Some notation was unexplained and formatting was inconsistent.	We reviewed the manuscript for undefined notation, inconsistent terminology, and formatting issues and corrected these throughout.	Throughout
C21	R1	The analysis is limited to a U.S. political context.	We agree that this is an important limitation of the present study. The revised Discussion states explicitly that the findings are based on U.S. candidate-centered political discussion during one election period and that comparable analyses in other national contexts are needed to establish how broadly the structural patterns generalize.	Limitations